

# GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

## SEMESTER IV

Name of Department: - Computer Science and Engineering

1. Subject Code: **TCS-403** Course Title: **Microprocessors**
2. Contact Hours: L: **3** **0** **0**
3. Examination Duration (Hrs): Theory **3** Practical **0**
4. Relative Weight: CIE **25** MSE **25** SEE **50**
5. Credits: **3**
6. Semester: **IV**
7. Category of Course: **DC**
8. Pre-requisite: TEC-101 Basic Electronics, TCS-301-Logic design

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| 9. Course Outcome**: | <p>After completion of the course the students will be able to:</p> <p><b>CO1:</b> Remember the concept of microcomputer system.</p> <p><b>CO2:</b> Understand microprocessor 8085, 8086 and microcontroller 8051 hardware.</p> <p><b>CO3:</b> Apply the concepts of assembly language programming of 8085 and 8086 to fulfil different tasks.</p> <p><b>CO4:</b> Examine the application of 8085 and 8086 microprocessor with interrupt system, real time timer and counter.</p> <p><b>CO5:</b> Test different interfacing ICs and memory for defined tasks with 8085 and 8086 microprocessor.</p> <p><b>CO6:</b> Integrate the knowledge of 8085, 8086 and 8051 in various embedded systems.</p> |
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**\*\* Describe the specific knowledge, skills or competencies the students are expected to acquire or demonstrate.**

### 10. Details of the Course:

| Sl. No. | Contents                                                                                                                                                                                                                         | Contact Hours |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1       | <b>Unit1: Microprocessor Architecture:</b> Introduction to Microprocessor and Microcomputer Architecture, Brief Evolution of Microprocessor, 8085 Register Organization, Internal architecture, Pins & Signals, 8085 Interrupts. | 10            |
| 2       | <b>Unit 2: Programming with 8085:</b> Programming model of 8085, 8085 instructions set, Addressing modes, Instruction Timing & Execution, Assembly language Programs using Instruction of 8085. Stack and subroutine.            | 10            |
| 3       | <b>Unit 3: Interfacing with 8085, Memory Interfacing:</b> Interfacing EPROM&RAM memories. Address Decoding, Memory & I/O mapped I/O,                                                                                             | 10            |

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|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|   | Programmable Interfaces: - 8255, 8251, 8253, 8259, Overview of DMA & DMA controller, ADC and DAC interfacing.                                                                                                                                                  |           |
| 4 | <b>Unit 4: Intel 8086 (16 bit processor):</b> 8086 Architecture, Physical address, segmentation, memory organization, Instruction set of 8086, Addressing Modes, Pins & Signals, 8086 Interrupts. Simple Assembly Language Programs using Instruction of 8086. | 10        |
| 5 | <b>Unit 5: 8051 Microcontroller:</b> Microprocessor Vs Microcontroller, Embedded Systems, 8051 Architecture- Registers, Pin diagram, I/O ports functions, Memory organization. Interrupts of 8051.                                                             | 08        |
|   | Total                                                                                                                                                                                                                                                          | <b>48</b> |

#### 11. Suggested Books:

| SL. No. | Name of Authors/Books/Publishers                                                                                                            | Edition                                                  | Year of Publication / Reprint |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------|
|         | <b>Textbooks</b>                                                                                                                            |                                                          |                               |
| 1.      | Ramesh Gaonkar, Microprocessor Architecture, Programming, and Applications with the 8085, Penram International Publication (India) Pvt. Ltd | 6 <sup>th</sup>                                          | 2013                          |
| 2       | A. K. Ray & K. M. Bhurchandi, Advanced Microprocessors and peripherals, Tata McGraw Hill                                                    | 3 <sup>rd</sup>                                          | 2012                          |
| 3.      | Muhammad Ali Mazidi, Janice Gillispie Mazidi, The 8051 Microcontroller & Embedded System, Pearson / PHI publication                         | 2 <sup>nd</sup>                                          | 2007                          |
|         | <b>Reference Books</b>                                                                                                                      |                                                          |                               |
| 1.      | Douglas V. Hall, Microprocessors and Interfacing, Tata McGraw Hill                                                                          | 3 <sup>rd</sup>                                          | 2012                          |
| 2.      | Barry B. Brey, The Intel Microprocessors Architecture Programming and interfacing, Pearson                                                  | 8 <sup>th</sup>                                          | 2012                          |
| 12.     | <b>Mode of Evaluation</b>                                                                                                                   | Test / Quiz / Assignment / Mid Term Exam / End Term Exam |                               |